EXPERIMENT 3: Create a Virtual Machine Using a Hypervisor and Install Linux

1. Aim

To create a virtual machine using a hypervisor and install a Linux operating system on it.

2. Procedure

1. Download and install Oracle VirtualBox or VMware Workstation on the host computer.
2. Download a Linux ISO image, such as Ubuntu, from its official website.
3. Open VirtualBox and click the New button.
4. Enter a virtual machine name such as Ubuntu and select Linux as the operating system type.
5. Allocate RAM according to the host computer's available memory; 4 GB is a typical lab allocation.
6. Create a virtual hard disk; allocate approximately 25 GB or more if available.
7. Attach the downloaded Linux ISO image to the virtual optical drive.
8. Start the virtual machine and select the Install Ubuntu option.
9. Choose the language, keyboard layout and installation type.
10. Configure the disk installation option and create a username and password.
11. Wait for the operating system installation to complete.
12. Restart the virtual machine, detach the ISO if prompted and log in using the created account.

3. Program / Commands

Not required. This experiment is performed using the specified software tools and commands.

4. Output of Program

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VirtualBox displays the installed Linux desktop inside the virtual machine.
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5. Result

Thus, a virtual machine was created using a hypervisor, and the Linux operating system was successfully installed and started.

EXPERIMENT 3: Create a Virtual Machine Using a Hypervisor and Install Linux

1. Aim

To create and configure a virtual machine using a hypervisor such as Oracle VirtualBox or VMware Workstation and install a Linux operating system in the virtual environment.

2. Procedure

1. Start the host computer and ensure that hardware virtualization is enabled in the BIOS/UEFI settings if required. 2. Download Oracle VirtualBox or VMware Workstation from its official website. 3. Install the virtualization software by following the installation wizard. 4. Download a Linux distribution ISO image, such as Ubuntu, from its official website. 5. Open the installed virtualization software and select the New option to create a virtual machine. 6. Enter a suitable name for the virtual machine, such as Ubuntu Linux. 7. Select Linux as the operating system type and choose the appropriate Ubuntu version. 8. Allocate sufficient RAM to the virtual machine based on the host computer's available memory. 9. Create a new virtual hard disk and allocate sufficient storage space, preferably around 25 GB or more. 10. Select the appropriate virtual hard disk format and choose dynamically allocated storage if required. 11. Open the virtual machine settings and attach the downloaded Linux ISO image to the virtual optical drive. 12. Start the virtual machine. The system will boot from the ISO image and display the Linux installation screen. 13. Select the language, keyboard layout and installation preferences. 14. Choose the installation option and configure the virtual disk according to the installation requirements. 15. Enter the required username, computer name and password. 16. Wait until the installation process is completed. The installer copies the required operating system files into the virtual hard disk. 17. Restart the virtual machine after installation and remove the installation ISO when prompted. 18. Log in using the previously created username and password. 19. Open the Linux desktop and verify that the operating system is functioning correctly. 20. Open the terminal and execute basic commands such as pwd, ls and uname -a to verify the installation.

4. Output of Program

Expected Output: - The VirtualBox or VMware window displays the newly created Linux virtual machine. - The Ubuntu login screen appears after installation. - After successful login, the Linux desktop environment is displayed. - Basic Linux commands execute successfully in the terminal. Sample Terminal Output: $ pwd /home/user $ uname -s Linux Explanation: The Linux operating system runs inside the virtual machine without replacing the host operating system. The virtual machine uses the hardware resources allocated by the hypervisor.

5. Result

Thus, a virtual machine was successfully created using a hypervisor, and the Linux operating system was installed and configured in the virtual environment. The installation was verified by accessing the Linux desktop and executing basic terminal commands.